Quartic normal equations: complete-space solvability, sharp inverse stability, and atomic obstructions

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Abstract

Let H be a real Banach space and let $\delta : H \to L^4(\mu)$ be bounded linear with $c\,\|u\| \le \|\delta u\|_4$ for some $c > 0$. The quartic normal map $\mathcal{A}(u)(v) = \int (\delta u)^3 \delta v \, d\mu$ is a bijection from H onto its continuous dual. We give a direct proof using norm-Cauchy minimizing sequences and obtain a global inverse estimate with exponent $1/3$. A weighted lower bound at a source gives a stronger linear stability estimate relative to that source. On infinite Poisson L^4, however, the exponent $1/3$ is locally optimal at every dual datum, including data whose inverse is nonzero almost everywhere. Every fixed finite-dimensional set of source directions can still admit linear stability at such a source. The results are reconstructed from a frozen Lean development; the existence and uniform-convexity mechanisms are classical. The purpose is a complete account of the estimates, their hypotheses, and the distinction between finite-direction and ambient regularity. Priority for the combined presentation remains unresolved.

1 The equation and its regularity questions

Fix a measure space (Ω, Σ, μ). No finiteness or σ-finiteness assumption is imposed unless explicitly stated. All L^p spaces in this paper consist of equivalence classes of real measurable functions modulo equality almost everywhere. Let H be a real Banach space with norm $\|\cdot\|$ and continuous dual H^*, equipped with its operator norm. Suppose that $\delta : H \to L^4(\mu)$ is bounded and linear and that

$$c\,\|u\| \le \|\delta u\|_4 \qquad (u \in H), \qquad c > 0. \tag{1}$$

Write $J(F) = F^3 \in L^{4/3}(\mu)$ and define

$$\mathcal{A}(u)(v) = \int_\Omega (\delta u)^3 \delta v \, d\mu, \qquad u, v \in H. \tag{2}$$

The notation $\delta^* J(\delta u)$ is permissible if δ^* means the transpose of δ applied to the integral functional represented by $J(\delta u)$. We do not identify all of $(L^4)^*$ with $L^{4/3}$ as a preliminary assumption. This matters on general measure spaces.

The lower bound in (1) is quantitative. Injectivity alone does not replace it in infinite dimension. It makes $u \mapsto \|\delta u\|_4$ an equivalent norm and makes the image of δ closed. Indeed, if δu_n converges in L^4, the lower bound makes u_n Cauchy, and completeness and continuity identify the limit. This elementary observation explains why the source norm is controlled throughout the paper.

Theorem 1.1. *Under* (1)*, the map* $\mathcal{A} : H \to H^*$ *is a smooth bijection. Its inverse* T *satisfies, for all* $f, g \in H^*$*,*

$$\|Tf - Tg\|^3 \le \frac{4}{c^4} \|f - g\| . \tag{3}$$

If H *contains a nonzero vector, no exponent* $q > 1/3$ *gives a local Hölder estimate for* T *at zero. If* $H = L^4(\mu_\lambda)$*,* δ *is the identity, and* μ_λ *is the Poisson law of a parameter* $\lambda > 0$*, no such exponent gives a local estimate at* any *datum of* H^**.*

The final assertion is stronger than failure of a uniform global Lipschitz constant. It excludes even an estimate comparing one fixed datum to nearby data. Conversely, we will prove such a fixed-datum linear estimate under an additional weighted lower bound. Its constant need not work between two arbitrary nearby data.

The existence argument belongs to the classical geometry of uniformly convex spaces, initiated by Clarkson [1]. Monotone-operator solvability provides another classical context [2, 3]. We do not claim a new general surjectivity principle or a new uniform-convexity inequality. The direct proof is useful here because it exposes the precise quantitative input and avoids a separate reflexivity construction. In classical analysis the closed embedding in L^4 already implies reflexivity; avoiding that theorem in the proof does not produce nonreflexive examples satisfying (1).

2 The actual duality map and its energy

Hölder's inequality gives

$$\|J(F)\|_{4/3} = \|F\|_4^3 , \qquad \int J(F) F \, d\mu = \|F\|_4^4 . \tag{4}$$

For example, $\int |F^3|^{4/3} = \int |F|^4 < \infty$, which proves membership and the norm identity directly. Almost-everywhere equality of two representatives implies the same equality for their cubes, so J is well defined on the quotient space. The product $J(F)G$ is integrable for every $G \in L^4$. Consequently (2) defines a continuous functional, with

$$\|\mathcal{A}(u)\| \le \|\delta\| \, \|\delta u\|_4^3 , \qquad \langle \mathcal{A}(u), u \rangle = \|\delta u\|_4^4 . \tag{5}$$

Multiplication is a bounded bilinear map $L^4 \times L^4 \to L^2$ and also $L^2 \times L^4 \to L^{4/3}$, both of norm at most one. The cube map is the composition of these maps along the diagonal. It is therefore a continuous cubic polynomial, in particular smooth. Composition with δ and the bounded transpose pairing proves smoothness of $\mathcal{A}$. This argument concerns Fréchet differentiability in the specified Banach norms; pointwise differentiation alone would not establish it.

For $f \in H^*$ put

$$E_f(u) = \frac{1}{4} \|\delta u\|_4^4 - f(u). \tag{6}$$

Lemma 2.1. *The energy is continuously Fréchet differentiable, and* $DE_f(u) = \mathcal{A}(u) - f$*.*

Proof. Write $F = \delta u$ and $G = \delta h$. Expansion of the quartic yields

$$\frac{1}{4} \int (F + G)^4 - \frac{1}{4} \int F^4 - \int F^3 G = \frac{3}{2} \int F^2 G^2 + \int F G^3 + \frac{1}{4} \int G^4 .$$

Every integral is absolutely convergent by Hölder's inequality. The absolute value of the right side is at most

$$\frac{3}{2} \|F\|_4^2 \, \|G\|_4^2 + \|F\|_4 \, \|G\|_4^3 + \frac{1}{4} \|G\|_4^4 .$$

Since $\|G\|_4 \le \|\delta\|\,\|h\|$, this is $o(\|h\|)$. The linear term is exactly $\mathcal{A}(u)(h)$. Smoothness of $\mathcal{A}$, established above, gives continuity of the derivative. □

Lemma 2.2. *The range of E_f is bounded below. More precisely, with $R = \max\{1, 4\|f\|/c^4\}$, one has $E_f(u) \ge -\|f\|\,R$.*

Proof. If $E_f(u) \le 0$ and $s = \|u\|$, then $c^4 s^4 \le 4\|f\|\,s$. For $s > 1$ this implies $c^4 s \le 4\|f\|$, hence $s \le R$; for $s \le 1$ the bound is immediate. Thus $E_f(u) \ge -f(u) \ge -\|f\|\,R$ on the zero sublevel. Outside that sublevel the same lower bound holds because the energy is positive and $\|f\|\,R \ge 0$. □

3 Minimization without weak compactness

The following elementary lemma isolates the whole existence mechanism. It will be applied after the required deficit has been proved for the actual quartic integral.

Lemma 3.1 (Quartic midpoint minimization)**.** *Let X be a real Banach space and let $E : X \to \mathbb{R}$ be continuous and bounded below. If $k > 0$ and*

$$E\left(\frac{x+y}{2}\right) + k\,\|x-y\|^4 \le \frac{E(x)+E(y)}{2} \qquad (x, y \in X), \tag{7}$$

then E attains its infimum.

Proof. Let $\alpha = \inf E(X)$, a finite real number. Choose x_n with $E(x_n) \to \alpha$. Since $\alpha \le E((x_n + x_m)/2)$, (7) implies

$$k\,\|x_n - x_m\|^4 \le \frac{E(x_n) - \alpha + E(x_m) - \alpha}{2}.$$

Given $\varepsilon > 0$, eventually both energy differences are less than $k\varepsilon^4$. The displayed inequality then gives $\|x_n - x_m\| < \varepsilon$. Completeness supplies a norm limit x, and continuity gives $E(x) = \alpha$. □

Lemma 3.2. *The energy* (6) *satisfies*

$$E_f\left(\frac{u+v}{2}\right) + \frac{c^4}{64}\,\|u-v\|^4 \le \frac{E_f(u)+E_f(v)}{2}. \tag{8}$$

Proof. For real a, b, direct expansion gives

$$\frac{a^4+b^4}{2} - \left(\frac{a+b}{2}\right)^4 - \frac{(a-b)^4}{16} = \frac{3}{8}(a^2-b^2)^2 \ge 0.$$

Substitute representatives of δu and δv and integrate. All fourth powers are integrable, so the resulting inequality is valid for the quotient classes. Multiplication by $1/4$ gives the deficit $\|\delta(u-v)\|_4^4/64$ for the potential. The forcing term cancels exactly at the midpoint by linearity of f. Finally apply (1) to $u - v$. □

Proposition 3.3. *For each $f \in H^*$ there exists $u \in H$ with $\mathcal{A}(u) = f$.*

Proof. Lemmas 2.2, 3.2, and 3.1 produce a global minimum u of E_f. For any $h \in H$, the real function $s \mapsto E_f(u + sh)$ has a minimum at zero, so its derivative there vanishes. Lemma 2.1 identifies that derivative with $(\mathcal{A}(u) - f)(h)$. Since this holds for every h, $\mathcal{A}(u) = f$. □

The proof uses neither compactness of a norm ball nor an unproved weakly convergent subsequence. The distinction is useful in a formal development: the only limiting step is completeness applied to a sequence already proved Cauchy. The argument also covers the zero space, where the equation has its unique trivial solution.

4 Uniform monotonicity and global inverse stability

Existence did not need a uniqueness assumption. The same scalar quartic geometry supplies uniqueness and the precise modulus of the inverse.

Lemma 4.1. *For $u, v \in H$,*

$$\langle \mathcal{A}(u) - \mathcal{A}(v), u - v \rangle \geq \frac{1}{4} \|\delta(u - v)\|_4^4 \geq \frac{c^4}{4} \|u - v\|^4 . \tag{9}$$

Proof. The scalar identity

$$(a^3 - b^3)(a - b) - \frac{1}{4}(a - b)^4 = \frac{3}{4}(a - b)^2(a + b)^2$$

has a nonnegative right side. Its terms are integrable after substituting $a = \delta u$, $b = \delta v$, by the $L^{4/3}$–L^4 pairing and quartic integrability. Integration gives the first inequality, and (1) gives the second. □

Proof of bijectivity and (3). If $\mathcal{A}(u) = \mathcal{A}(v)$, (9) forces $u = v$. Together with Proposition 3.3, this defines the inverse $T : H^* \to H$. For $u = Tf$, $v = Tg$, dual norm control gives

$$\frac{c^4}{4} \|u - v\|^4 \leq (f - g)(u - v) \leq \|f - g\| \, \|u - v\| .$$

When $u \neq v$, divide by $\|u - v\|$; otherwise the desired estimate is immediate. Taking cube roots gives the global Hölder bound

$$\|Tf - Tg\| \leq (4/c^4)^{1/3} \, \|f - g\|^{1/3} .$$

□

Proposition 4.2. *Suppose $u \neq 0$. For every $q > 1/3$, $K \geq 0$, and $r > 0$, there exists $f \in H^*$ with $\|f\| < r$ and*

$$\|Tf - T0\| > K \, \|f\|^q .$$

Proof. The defining integral gives $\mathcal{A}(au) = a^3 \mathcal{A}(u)$ for real a and $\mathcal{A}(0) = 0$. Put $f_a = \mathcal{A}(au)$ for $a > 0$. Then $Tf_a = au$, $T0 = 0$, and $\|f_a\| = a^3 \, \|\mathcal{A}(u)\| \to 0$. The proposed estimate would require

$$\|u\| \leq K \, \|\mathcal{A}(u)\|^q \, a^{3q-1}.$$

The right side tends to zero, whereas $\|u\| > 0$. Choosing a small enough gives the strict reverse inequality and $\|f_a\| < r$. □

The exponent is optimal in this precise local sense; no claim that the global constant is optimal for every δ is needed. The singularity at zero follows from cubic homogeneity. To understand nonzero data we need information about multiplication by the source.

5 Weighted source control and finite-dimensional directions

For $u \in H$, define the bounded linear map

$$\mathcal{B}_u : H \to L^2(\mu), \qquad \mathcal{B}_u w = (\delta u)(\delta w). \tag{10}$$

Hölder's inequality gives $\|\mathcal{B}_u w\|_2 \le \|\delta u\|_4 \, \|\delta\| \, \|w\|$. This is an actual L^2 product, with equality of representatives understood almost everywhere.

Lemma 5.1. *For all $u, v \in H$,*

$$\langle \mathcal{A}(u) - \mathcal{A}(v), u - v \rangle \ge \frac{3}{4} \, \|\mathcal{B}_u(u - v)\|_2^2 . \tag{11}$$

Proof. For real a, b,

$$(a^3 - b^3)(a - b) - \frac{3}{4} a^2 (a - b)^2 = (a - b)^2 (b + a/2)^2 \ge 0.$$

Integration gives the assertion. The square of the product is integrable because $(\delta u)(\delta(u - v)) \in L^2$. □

Theorem 5.2 (Stability relative to one datum)**.** *If $\beta > 0$ and $\|\mathcal{B}_u w\|_2 \ge \beta \, \|w\|$ for every $w \in H$, then, for every $v \in H$,*

$$\|u - v\| \le \frac{4}{3\beta^2} \, \|\mathcal{A}(u) - \mathcal{A}(v)\| . \tag{12}$$

Under (1)*, setting $f_0 = \mathcal{A}(u)$ yields the corresponding bound $\|T f_0 - T f\| \le 4 \, \|f_0 - f\| / (3\beta^2)$ for all $f \in H^*$.*

Proof. By Lemma 5.1 and the dual norm inequality,

$$\frac{3}{4} \beta^2 \, \|u - v\|^2 \le \|\mathcal{A}(u) - \mathcal{A}(v)\| \, \|u - v\| .$$

Divide by the distance when it is positive. The zero case is immediate. □

Proposition 5.3. *Suppose δ is injective and δu is nonzero almost everywhere. For each linear subspace $E \subset H$ of finite dimension, there is $K_E > 0$ such that*

$$\|u - v\| \le K_E \, \|\mathcal{A}(u) - \mathcal{A}(v)\| \qquad \text{whenever } u - v \in E. \tag{13}$$

In particular, if H is finite dimensional the estimate holds for every $v \in H$.

Proof. If $\mathcal{B}_u w = 0$, then $(\delta u)(\delta w) = 0$ almost everywhere. Nonvanishing of δu gives $\delta w = 0$, and injectivity gives $w = 0$. Thus $\mathcal{B}_u$ is injective. Restricted to E, it has a positive lower norm bound: for $E \neq \{0\}$, minimize the continuous positive function $w \mapsto \|\mathcal{B}_u w\|_2$ on the compact unit sphere of E. Its minimum β_E is positive; scaling gives the bound on E. For $E = \{0\}$ any positive lower-bound constant works. Apply the proof of Theorem 5.2 to $w = u - v \in E$. □

The finite-dimensional constant depends on the subspace. The proposition provides no uniform lower bound as E varies. This dependence is decisive on an infinite atomic space.

As a regular example, let μ be a probability measure, $H = \mathbb{R}$, and δu the constant function u. Then $\|\delta u\|_4 = |u|$ and $\|\mathcal{B}_u w\|_2 = |u||w|$. At $u \neq 0$, (12) gives a linear stability constant $4/(3|u|^2)$. Under the usual identification of $\mathbb{R}^*$ with $\mathbb{R}$, the normal equation is $u^3 = f$. The behavior at zero and at nonzero data is thus already different in one dimension.

6 Infinite Poisson space and failure of weighted coercivity

Fix $\lambda > 0$. Give $\mathbb{N} = \{0, 1, \ldots\}$ its full power-set σ-algebra and Poisson probability measure

$$p_n = \mu_\lambda(\{n\}) = e^{-\lambda}\frac{\lambda^n}{n!} > 0.$$

In this section $H = L^4(\mu_\lambda)$ and δ is the identity. The lower bound (1) holds with $c = 1$. Every coordinate has positive mass, so an equivalence class has well-defined values $F(n)$. Its norm is

$$\|F\|_4^4 = \sum_{n=0}^{\infty} p_n |F(n)|^4.$$

Let $e_n = p_n^{-1/4} 1_{\{n\}}$. These are normalized in L^4. For a fixed $F \in H$ put $x_n = F(n)p_n^{1/4}$. Summability gives

$$\sum_n |x_n|^4 < \infty, \qquad x_n \longrightarrow 0. \tag{14}$$

Proposition 6.1. *No $F \in L^4(\mu_\lambda)$ satisfies a positive lower bound $\|Fw\|_2 \geq \beta \|w\|_4$ for all $w \in L^4(\mu_\lambda)$.*

Proof. Taking $w = e_n$ gives

$$\|Fe_n\|_2 = |F(n)|p_n^{1/4} = |x_n| \longrightarrow 0, \qquad \|e_n\|_4 = 1.$$

A positive lower bound contradicts this limit. □

Even the constant source $F = 1$ has this obstruction. It is nonzero everywhere, and multiplication by F is injective, but the inclusion of this probability-space L^4 into L^2 has no positive lower bound. This does not yet prove failure of inverse Lipschitz stability: Theorem 5.2 is a sufficient criterion, not an equivalence. We next prove the stronger failure directly by perturbing the normal map itself.

7 The inverse exponent is optimal at every atomic datum

The integral pairing represented by e_n^3 has dual norm at most one by Hölder's inequality. For $a \geq 0$, the functions $F + ae_n$ and F differ at only one coordinate. Cubing gives the exact identity

$$\mathcal{A}(F + ae_n) - \mathcal{A}(F) = ((x_n + a)^3 - x_n^3)\mathcal{A}(e_n). \tag{15}$$

In particular,

$$\|\mathcal{A}(F + ae_n) - \mathcal{A}(F)\| \leq a(3|x_n|^2 + 3a|x_n| + a^2). \tag{16}$$

To check (15) without confusing values and normalized coordinates, observe that on the nth atom $F(n) = p_n^{-1/4}x_n$ and $e_n(n) = p_n^{-1/4}$. The cube difference is therefore $p_n^{-3/4}((x_n+a)^3 - x_n^3)$, exactly the multiplier of e_n^3. Off that atom both sides vanish.

Theorem 7.1. *For every $f_0 \in H^*$, $q > 1/3$, $K \geq 0$, and $r > 0$, there exists $f \in H^*$ such that*

$$\|f - f_0\| < r, \qquad \|Tf - Tf_0\| > K \|f - f_0\|^q. \tag{17}$$

Proof. By surjectivity write $F = Tf_0$. Let

$$a_n = |x_n| + \frac{1}{n+1}, \qquad U_n = F + a_n e_n, \qquad f_n = \mathcal{A}(U_n).$$

Then $a_n > 0$ and $a_n \to 0$ by (14). Since $|x_n| \le a_n$, (16) gives

$$\|f_n - f_0\| \le 7a_n^3 \longrightarrow 0.$$

Injectivity of $\mathcal{A}$ and $\|e_n\|_4 = 1$ give $\|Tf_n - Tf_0\| = \|U_n - F\|_4 = a_n$. Because $3q - 1 > 0$, eventually $K7^q a_n^{3q-1} < 1$. Hence

$$K\,\|f_n - f_0\|^q \le K7^q a_n^{3q} < a_n.$$

Choose n large enough for this inequality and the radius condition. □

This completes the optimality assertion in Theorem 1.1. The proof uses small normalized coordinates of an arbitrary L^4 source, rather than a zero coordinate. Thus it applies to strictly positive sources as well as sources supported on finitely many atoms. The Poisson distribution is a concrete verified model; the argument itself only uses the displayed positive atomic weights and quartic summability. We do not claim a separate Lean theorem for every atomic measure under that generalization.

Corollary 7.2. *At the constant source $F = 1$, every fixed finite-dimensional source subspace $E \subset L^4(\mu_\lambda)$ admits* (13). *There is no common constant valid for all one-dimensional source subspaces.*

Proof. Apply Proposition 5.3 to $\delta = \mathrm{id}$ and $F = 1$. If a constant K worked for all one-dimensional subspaces, then for each V take $E = \mathrm{span}\{1 - V\}$. The estimate would hold for every V. Taking $V = Tf$ contradicts Theorem 7.1 with $f_0 = \mathcal{A}(1)$ and $q = 1$. □

The order of quantifiers is the issue: for each fixed E there is a constant, but no single constant works for all E. A finite-dimensional calculation at a nonvanishing source can therefore be accurate while providing no uniform ambient regularity theorem.

8 Disjoint sources and the role of support

A simpler obstruction at selected nonzero data helps separate two mechanisms. Assume (1). Suppose $u, v \in H$, $v \neq 0$, and $(\delta u)(\delta v) = 0$ almost everywhere. Then mixed terms in the cubic vanish, so

$$\mathcal{A}(u + av) = \mathcal{A}(u) + a^3\mathcal{A}(v) \qquad (a \in \mathbb{R}). \tag{18}$$

Proposition 8.1. *For every $K \ge 0$ and $r > 0$ there exists f with $\|f - \mathcal{A}(u)\| < r$ and $\|Tf - u\| > K\,\|f - \mathcal{A}(u)\|$.*

Proof. Use $f_a = \mathcal{A}(u + av)$ for $a > 0$. The source difference has norm $a\,\|v\|$, and (18) gives dual difference $a^3\,\|\mathcal{A}(v)\|$. Small a meets both required inequalities. □

For the Poisson identity source one may take indicators of two distinct atoms. This produces a nonzero datum with no linear inverse estimate using disjoint support alone. Theorem 7.1 goes further: it works when the source never vanishes and no nonzero disjoint source exists. Small normalized tail coordinates, rather than exact support separation, then replace cubic decoupling.

9 Formal scope and reproducibility

The mathematical baseline is the frozen Git commit

`bab688bb6140cd6316c80680095a914ea0d346db`.

All declarations below have prefix `OperatorDerivativeRewrite.Foundation`. The accompanying `theorem-map.json` records exact files, declarations, and hashes. The principal correspondence is as follows.

Paper result	*Lean namespace and declaration*
(4), (2)	`BanachLpDualityFour.dualityFour_norm`, `dualityFour_pairing_eq_norm`, `normalFour`
Lemma 2.1	`BanachLpDualityFourExistence.hasFDerivAt_quarticPotential`
Lemmas 2.2–3.2	`BanachLpDualityFourCoercivity.quarticPotential_lower_bound`, `quarticEnergy_midpoint_lower`; `QuarticUniformMinimization.exists_minimizer_of_quartic_midpoint`
Bijectivity, (3)	`BanachLpDualityFourInfinite.normalFour_existsUnique`, `normalFour_inverse_holder`
Proposition 4.2	`BanachQuarticHolderSharpness.no_local_holder_bound`
Theorem 5.2	`BanachQuarticWeightedStability.inverse_lipschitz_at_weighted_datum`
Proposition 5.3	`BanachFiniteDirectionStability.finite_direction_stability`
Proposition 6.1	`PoissonQuarticWeightedObstruction.no_positive_weighted_lower`
(15), (16)	`PoissonQuarticAtomPerturbation.normal_atom_shift`, `normal_shift_norm_bound`
Theorem 7.1	`PoissonQuarticHolderEverywhere.no_local_holder_at_datum`
Corollary 7.2	`BanachFiniteDirectionStability.poisson_constant_finite_directions`, `no_uniform_one_direction_bound`
Proposition 8.1	`BanachQuarticDisjointObstruction.no_local_linear_bound_at_disjoint_image`

The complete-space namespace is essential in the existence row: an earlier namespace contains only a finite-dimensional existence theorem, although its energy derivative is dimension independent. The map called `normalFourEquiv` is a nonlinear bijection, not a linear equivalence or an identification supplied by a Hilbert inner product. The paper's H^* is the continuous real dual.

The baseline audit receipt records a successful complete root build, 54,943 audited declarations, and acceptance only of `propext`, `Classical.choice`, and `Quot.sound`. The editorial work changes no Lean source or dependency configuration. It reuses that existing verification rather than claiming a new full build. The source hashes and receipt provenance are included with the manuscript. Lean is pinned to version 4.34.1 and Mathlib to revision `d13f23b723b8a846827a245b89c10fc7d3f11612`.

10 What the estimates do and do not decide

The lower bound on δ gives existence on the whole continuous dual, uniqueness, and a global modulus. A weighted lower bound gives stronger control relative to one source. Nonvanishing

almost everywhere implies injectivity of the weighted map; finite dimension converts that injectivity into a lower bound. Infinite Poisson space shows exactly where that last implication fails, and direct perturbations show that the failure corresponds to an actual inverse singularity.

No stochastic integral is constructed by these arguments. In particular, a Banach-space lower bound for a proposed stochastic divergence cannot be presumed from the existence theorem: it is an input that must be established for the actual operator. The results also do not settle general exponents, unbounded transpose domains, or closability characterizations. These are separate questions.

The solvability and uniform monotonicity belong to established nonlinear analysis. The bounded preliminary literature check places them in that context and does not certify priority for the particular combination of atomic optimality and finite-direction regularity. This manuscript is a reviewed exposition of verified mathematics, with mathematical novelty explicitly unresolved. An external submission would require a more targeted comparison with the literature on inverse duality mappings and their local moduli.

References

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